Early Giants: Black-Hole Growth Constraints from JWST

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Abstract

Aims. Massive black holes in the early Universe had limited time to assemble. We identify which JWST black-hole candidates place the strongest constraints on early growth models and assess how these constraints depend on mass estimates and growth assumptions. **Methods.** We compile 355 published measurements at $z \geq 4$ and analyse a primary sample of 223 objects with broad Balmer-line masses. An expanded sample adds 14 tentative or conditional Balmer-line candidates and GN-z11, whose mass is inferred from ultraviolet lines. We propagate reported mass uncertainties and test mass-scale shifts and alternative growth assumptions. A focused primordial-seed comparison tests earlier accretion onset for the three highest-ranked primary objects and GN-z11. **Results.** For $100\,M_\odot$ seeds formed at $z = 30$, radiative efficiency $\epsilon = 0.1$, and no merger boost, 11 of the 223 objects require a lifetime-average Eddington ratio $\overline{f}_{\rm Edd,req} > 1$. Eight exceed unity in at least 95% of Monte Carlo draws using the reported mass uncertainties. For CAPERS-LRD-z9, CANUCS-LRD-z8.6, UNCOVER-20466, COSMOS3D-13852, and RUBIES-EGS-55604, at least 95% of draws still require an average Eddington ratio above 1 after reducing each mass by 0.5 dex (a factor of approximately 3.2). Delaying seeding to $z = 20$ gives GN-z11 the highest required average Eddington ratio. At $\epsilon = 0.05719$, the ideal thin-disc efficiency for a non-spinning black hole, all 238 central mass estimates are reachable from $100\,M_\odot$ seeds formed at $z = 30$ without lifetime-average accretion above the Eddington limit. At $\epsilon = 0.1$ and an average Eddington ratio capped at 1, accretion beginning at matter–radiation equality still requires minimum seeds of approximately 270–1200 $M_\odot$ for the four focused targets.

Key words. black hole physics – accretion, accretion disks – galaxies: active – galaxies: high-redshift – quasars: supermassive black holes

1 Introduction

The masses and redshifts of early black holes constrain how much growth is possible from an assumed seed. JWST surveys supply broad-line mass estimates across different selection methods and observing fields (Y. Harikane et al. 2023; J. Matthee et al. 2024; C. J. Baccus & X. Xu 2026). Growth studies of individual objects, including UNCOVER-20466 and GN-z11, compare light seeds, heavy seeds, and rapid accretion (V. Kokorev et al. 2023; R. Maiolino et al. 2024). Comparing these objects across studies requires consistent growth assumptions and explicit treatment of alternative measurements, uncertain identities, and mass-estimator errors.

We compile published measurements with their source versions, mass methods, and uncertainties. We retain alternative measurements and define primary and expanded samples by evidence for broad-line emission and the interpretation of the mass estimator. Readers can trace adopted masses to their sources and repeat the growth comparison with different measurements or sample choices.

We use the exponential growth relation applied by P. Dayal (2024) to rank required lifetime-average Eddington ratios. We test reported mass errors, separate calibration scatter, fixed mass shifts, seed masses, growth onset, and radiative efficiency. For the three highest-ranked primary objects and GN-z11, an earlier-onset comparison tests how available growth time changes the required seed mass, without modelling seed formation or gas supply.

2 Catalogue construction

We compile 355 literature measurements from 37 source families (studies or related surveys) at $z \geq 4$, with a literature cutoff of 3 September 2026. The catalogue is a curated collection of reproducibly extractable published measurements, not a statistically complete census of high-redshift black holes. Its scope does not include a systematic compilation of legacy ground-selected quasar samples. The bounded membership revision admitted on 17 October 2026 preserves that cutoff and the preceding edition at repository commit `fc0c5501ad22`. It adds CAPERS-LRD-z9 (Taylor et al. 2025), CANUCS-LRD-z8.6 (Tripodi et al. 2025), J1007_AGN (Schindler et al. 2025), MACS0416-Y1 (Takechi et al. 2026), and SAPPHIRES-ID-8760 (Fudamoto et al. 2025). The first four are Balmer source families in v2 and v3; the mass-free SAPPHIRES candidate is v3 only. Y1 remains candidate evidence in the expanded sample; its D2 component is one nucleus in one host. The source's mass is already lens-corrected, as is CANUCS's mass. SAPPHIRES has no published black-hole mass, so numerical growth inference is disabled; its weak Hbeta and unidentified foreground contaminant remain evidence caveats. Morishita's revised CANUCS analysis (Morishita et al. 2026) provides same-object evidence and reuses Tripodi's mass.

For J1007_AGN we adopt the observed Hbeta mass $(1.152^{+1.011}_{-0.463}) \times 10^8 M_\odot$ as a conservative comparable baseline, while the authors favor a dust-attenuated interpretation. Their correlated dust-corrected alternative is $(4.514^{+5.218}_{-2.320}) \times 10^8 M_\odot$; it raises the reference requirement from 1.113 to 1.222 without adding another object. CAPERS uses its source-canonical 7.58 ± 0.15 dex mass. Its 6.65–8.50 systematic envelope combines different assumptions, including an assumed Eddington luminosity bound, and is not sampled as a measured interval.

We adopt one measurement per object and retain alternatives, recording the source, method, uncertainty, publication version, and reason for each choice (Appendix A.2).

Table 1: Catalogue contents and analysis samples. Measurements are literature records of individual objects; an object may have records from several studies. Object counts are provisional where identifications across studies remain uncertain.

Category	Count
Measurements / provisional objects / host systems	355 / 343 / 342
Growth-eligible measurements / objects	248 / 241
Preliminary Balmer selection: measurements / objects	237 / 230
Primary sample / expanded sample	223 / 238
Objects with mass uncertainties (primary / expanded)	211 / 226
Objects excluded for uncertain identity	6
Objects without growth-eligible measurements	102

The preliminary Balmer selection requires a non-conditional Balmer virial mass and secure or probable evidence for an accreting black hole in the catalogue. These counts precede the final checks of individual line detections and object identities. Of these 230 objects, 4 tentative JADES detections enter only the expanded sample and 3 are excluded for uncertain identity, leaving 223 primary objects. The 102 objects without eligible masses include 3 of the 6 identity exclusions. Figure 5 counts those 3 under identity exclusions, leaving 99 in its bar for objects without eligible masses.

Growth analysis requires a numerical mass and redshift, an identified mass method, and a documented lensing correction where needed. Upper limits, indirect model ranges, and masses inferred by assuming an Eddington ratio are excluded from numerical inference. Entries with unresolved identities across studies are also excluded (Appendix A.2).

For the primary sample, we use masses estimated from broad Hα or Hβ emission lines in objects with secure or probable evidence for an accreting black hole, using the evidence classifications recorded from the source papers. Candidates with tentative broad-line detections or masses conditional on a broad-line-region interpretation enter the expanded sample only. Evidence for an active galactic nucleus (AGN) or a

broad-line region determines sample membership; potential biases in an accepted virial mass are assessed separately. COSMOS3D-13852 therefore remains in the primary sample on the basis of its published broad-line AGN identification; we test possible scattering bias in Section 5.1.

These single-epoch virial estimators combine the line width, which estimates gas velocity, with a luminosity-based estimate of the broad-line region's radius. Their calibration is based on reverberation mapping of nearby AGN (Y. Shen 2013). Using the same line family and mass-estimation framework reduces differences between methods in the comparison. Applying these calibrations at high redshift still requires assessing gas dynamics, geometry, obscuration, and lensing for each object.

The expanded sample adds 7 candidates with masses conditional on a broad-line-region interpretation, 7 with tentative broad-line detections, and GN-z11 to the primary sample (Appendix A.1). GN-z11's mass is inferred from rest-frame ultraviolet emission lines using local virial relations (R. Maiolino et al. 2024); its lines and calibration differ from those of the Balmer-line sample. These criteria yield 223 primary objects and 238 expanded-sample objects.

Objects without reported mass errors enter calculations at their central masses but not the uncertainty analysis. The mass estimates retained in each sample still use different calibrations, and the contributing studies have different target-selection criteria and detection limits. Appendix A gives the source inventory, identity decisions, and sample accounting. Table 1 summarizes the catalogue and analysis sample sizes.

3 Growth-model comparison

3.1 Theoretical framework

We use the luminosity-based Eddington ratio $f_{\rm Edd} = L_{\rm bol}/L_{\rm Edd}$, where $L_{\rm bol}$ is the bolometric luminosity of the accretion flow and $L_{\rm Edd}$ is the Eddington luminosity for electron scattering in ionized hydrogen. The fixed radiative efficiency ϵ is the fraction of inflowing rest-mass energy emitted as radiation; the fraction $1-\epsilon$ is retained by the black hole in this model. We define the available growth interval Δt and the lifetime-average Eddington ratio $\overline{f}_{\rm Edd}$ as

$$\Delta t = t(z_{\rm obs}) - t(z_{\rm seed}) > 0, \qquad \overline{f}_{\rm Edd} = \frac{1}{\Delta t}\int_{t(z_{\rm seed})}^{t(z_{\rm obs})} f_{\rm Edd}(t)\,dt. \tag{1}$$

Here, $t(z)$ is the age of the Universe at redshift z, $z_{\rm seed}$ is the redshift at which the seed forms and growth begins, and $z_{\rm obs}$ is the observed redshift. The average includes active and inactive phases. We use the exponential growth relation adopted by P. Dayal (2024), expressed in terms of the time-averaged Eddington ratio, and introduce a multiplier $B_{\rm merge}$ to approximate additional mass from mergers:

$$M_{\rm BH}(t_{\rm obs}) = B_{\rm merge} M_{\rm seed} \exp\left[\frac{1-\epsilon}{\epsilon}\frac{\overline{f}_{\rm Edd}\,\Delta t}{t_{\rm Edd}}\right], \tag{2}$$

Here, $M_{\rm BH}(t_{\rm obs})$ is the black-hole mass at the observation time $t_{\rm obs} = t(z_{\rm obs})$, and $M_{\rm seed}$ is the initial seed mass. The Eddington timescale is $t_{\rm Edd} = M_{\rm BH}c^2/L_{\rm Edd} \simeq 0.45$ Gyr, where c is the speed of light. The mass-growth e-folding time is 0.05 Gyr at $\epsilon = 0.1$ and $f_{\rm Edd} = 1$ (Appendix B). The dimensionless merger multiplier is $B_{\rm merge} = 1$ for the reference calculation and 2 for a factor-of-two mass boost. We represent mergers by multiplying the seed mass by a constant factor before calculating accretion. This approximation omits the timing and sequence of mergers, their mass ratios, and mass lost through gravitational radiation. The required lifetime-average ratio $\overline{f}_{\rm Edd,req}$ is the average needed to reach the observed mass $M_{\rm BH}$ from the assumed seed within Δt:

$$\overline{f}_{\rm Edd,req} = \max\left[0, \frac{\epsilon}{1-\epsilon}\frac{t_{\rm Edd}}{\Delta t}\ln\left(\frac{M_{\rm BH}}{B_{\rm merge}M_{\rm seed}}\right)\right]. \tag{3}$$

If $B_{\rm merge}M_{\rm seed}$ exceeds the observed mass, we set the required ratio to zero, but the seed and merger contribution already overpredict the mass. Such a history cannot reproduce the observation without mass loss. Inferring a duty cycle also requires assumptions about the rates during active and inactive phases.

We calculate cosmic ages in a flat cosmology with $H_0 = 67.3\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_m = 0.315$, and $\Omega_\Lambda = 0.685$, neglecting radiation. The first two values follow P. Dayal (2024). Appendix B gives the derivations, implementation, and inverse seed-mass relation. Equations 2–3 assume constant radiative efficiency. The ideal thin-disc efficiencies used in Figure 2 are 0.03775, 0.05719, and 0.42265 for dimensionless spins $a = -1, 0, +1$, respectively (J. M. Bardeen et al. 1972). Negative and positive spins are counter-aligned and aligned with the disc, respectively. Each track holds efficiency constant; spin evolution is unspecified.

3.2 Reference model

We calculate the required lifetime-average Eddington ratio using Eq. 3, with $M_{\rm seed} = 10^2 M_\odot$, $z_{\rm seed} = 30$, $\epsilon = 0.1$, and $B_{\rm merge} = 1$ as the reference model. We adopt $z_{\rm seed} = 30$ as an early starting point that allows more time for light-seed growth, and test how later onset changes the required accretion rates. We rank objects by this required ratio.

3.3 Uncertainty and sensitivity

We use Monte Carlo sampling to propagate reported asymmetric log-mass errors, with 10,000 draws per object. Each draw has equal probability of falling above or below the central mass, with its distance drawn from a half-normal distribution scaled by the corresponding upper or lower error. The central value is thus treated as a median. We use the quoted errors as one-sigma Gaussian scales, including when their confidence level is unspecified; we do not rescale them to a common confidence level. This is an approximation to the reported intervals, not a reconstruction of the source posterior. Redshifts are held fixed. For each sampled mass, we calculate the required average Eddington ratio and report the fraction of draws above unity, denoted P. Thus $P \geq 0.95$ means that at least 95% of draws exceed unity under this error model and the fixed growth assumptions. It is not a uniformly calibrated confidence statement across the sample. Quoted errors also differ in whether they include calibration scatter. Probability calculations use objects with reported mass uncertainties (Table 1).

We test mass-scale sensitivity by shifting all adopted log masses and their error draws by offsets of -0.5, -0.3, 0, $+0.3$, and $+0.5$ dex, keeping the sample and growth assumptions fixed. These offsets test shifts comparable to the uncertainty of single-epoch mass estimates (Y. Shen 2013). For each offset, we report the fraction of shifted Monte Carlo draws that require an average Eddington ratio above 1 under the reference model. Sampling and conversion details are given in Appendix B.5.

For RUBIES-EGS-55604 and J0910_2028_12910, we separately treat the reported 0.5-dex calibration scale as the standard deviation of an independent, zero-mean Gaussian error in log mass. Their reported errors are symmetric, so the combined standard deviation is $\sigma_{\rm tot} = \sqrt{\sigma^2_{\rm reported} + 0.5^2}$ dex. We evaluate the resulting normal tail above the mass required for $\overline{f}_{\rm Edd,req} = 1$ exactly, at offsets of 0 and -0.5 dex. This assumption tests scatter, separately from a fixed mass bias. We add no scatter to GN-38509 or ZS7, whose reported errors already include it, and assign none to objects without a separately quantified scale. This is a two-object sensitivity test, not a common error model for the full sample.

4 Results

4.1 Required accretion under the reference assumptions

For a $10^2\,M_\odot$ seed under the reference model, the median required lifetime-average Eddington ratio across the 223 primary objects is $\overline{f}_{\rm Edd,req} = 0.574$ (interquartile range 0.487–0.679), with eleven objects above unity and none above 2. Across the expanded 238-object sample, the corresponding median is 0.581 (interquartile range 0.488–0.689), with 17 above unity. These statistics use the required ratios at the adopted central masses. CAPERS-LRD-z9 has the highest required ratio, 1.519, with a 16th–84th percentile interval of 1.479–1.561 from its mass-error draws. UNCOVER-20466 (1.452) and CANUCS-LRD-z8.6 (1.448) follow. Table 2 lists the eleven primary objects above the threshold, ranked by the required ratios at their central masses.

The shorter growth time at $z = 10.603$ places GN-z11 fourth in the expanded-sample required-rate ranking despite its smaller mass.

Table 2: The eleven primary objects above the reference threshold, ordered by f_0. Redshifts are the adopted spectroscopic values. Masses and errors are in $\log_{10}(M_{\rm BH}/M_\odot)$; all estimators are single-epoch virial. f_0 is the required lifetime-average Eddington ratio evaluated at the adopted central mass under the reference model; f_{16} is the 16th percentile of this required ratio over the object's mass-error draws, and $P_0 = P(\overline{f}_{\rm Edd,req} > 1)$. f_- and P_- apply the fixed -0.5 dex offset. Δm (dex) is the reduction in log mass that brings the central required average Eddington ratio to 1, with the growth assumptions unchanged. All probabilities use the quoted mass errors and fixed growth assumptions. Additional mass-estimator limitations are in Table 7.

Object/source	z	Line	Log mass	f_0	f_{16}	P_0	f_-	P_-	Δm
CAPERS-LRD-z9 Taylor et al. (2025)	9.288	Hβ	$7.58^{+0.15}_{-0.15}$	1.519	1.479	> 0.999	1.383	> 0.999	1.907
UNCOVER-20466 J. E. Greene et al. (2024)	8.5	Hβ	$8.17^{+0.42}_{-0.42}$	1.452	1.354	> 0.999	1.335	> 0.999	1.921
CANUCS-LRD-z8.6 Tripodi et al. (2025)	8.6319	Hβ	$8.00^{+0.20}_{-0.22}$	1.448	1.394	> 0.999	1.327	> 0.999	1.856
COSMOS3D-13852 X. Lin et al. (2026)	7.646	Hβ	$8.60^{+0.35}_{-0.33}$	1.314	1.247	> 0.999	1.214	> 0.999	1.576
RUBIES-EGS-55604[a] D. D. Kocevski et al. (2025)	6.986	Hβ	$9.31^{+0.10}_{-0.10}$	1.267	1.250	> 0.999	1.180	> 0.999	1.541
CEERS-1019 R. L. Larson et al. (2023)	8.679	Hβ	$6.95^{+0.37}_{-0.37}$	1.205	1.115	0.988	1.083	0.822	0.842
J1007_AGN[d] Schindler et al. (2025)	7.2583	Hβ	$8.06^{+0.27}_{-0.22}$	1.113	1.072	0.997	1.022	0.702	0.618
GN-38509[b] I. Juodžbalis et al. (2026a)	6.678	Hα	$8.57^{+0.37}_{-0.38}$	1.065	1.005	0.860	0.984	0.392	0.398
J0910_2028_12910[a] C. J. Baccus & X. Xu (2026)	6.6214	Hα	$8.58^{+0.01}_{-0.01}$	1.053	1.051	> 0.999	0.973	< 0.001	0.330
CEERS-00717 Y. Harikane et al. (2023)	6.936	Hα	$7.99^{+0.16}_{-0.17}$	1.027	0.998	0.821	0.942	0.018	0.160
ZS7[c] H. Übler et al. (2024)	7.15	Hβ	$7.70^{+0.40}_{-0.40}$	1.023	0.954	0.625	0.934	0.173	0.130

[a] A separate 0.5-dex calibration scale is reported for RUBIES-EGS-55604 and J0910_2028_12910; these scales are omitted from the quoted errors and probabilities shown here. [b] GN-38509's quoted mass errors already include 0.3-dex calibration scatter. [c] ZS7's quoted 0.4-dex error already includes calibration scatter. [d] J1007_AGN has a separate 0.43-dex calibration scale, retained outside the quoted posterior errors. Unmarked rows have no separate scalar numerical calibration scale recorded; their quoted errors may omit other mass-estimator uncertainties.

4.2 Sensitivity to mass estimates

Of the eleven primary objects above unity at their central masses, nine remain above it at the 16th percentile and eight have $P \geq 0.95$ (Fig. 1). These results use the quoted mass errors, which include different combinations of measurement and calibration uncertainty (Table 2). Probability calculations use the 211 primary objects with reported errors. Appendix A.3 gives the expanded-sample percentile counts.

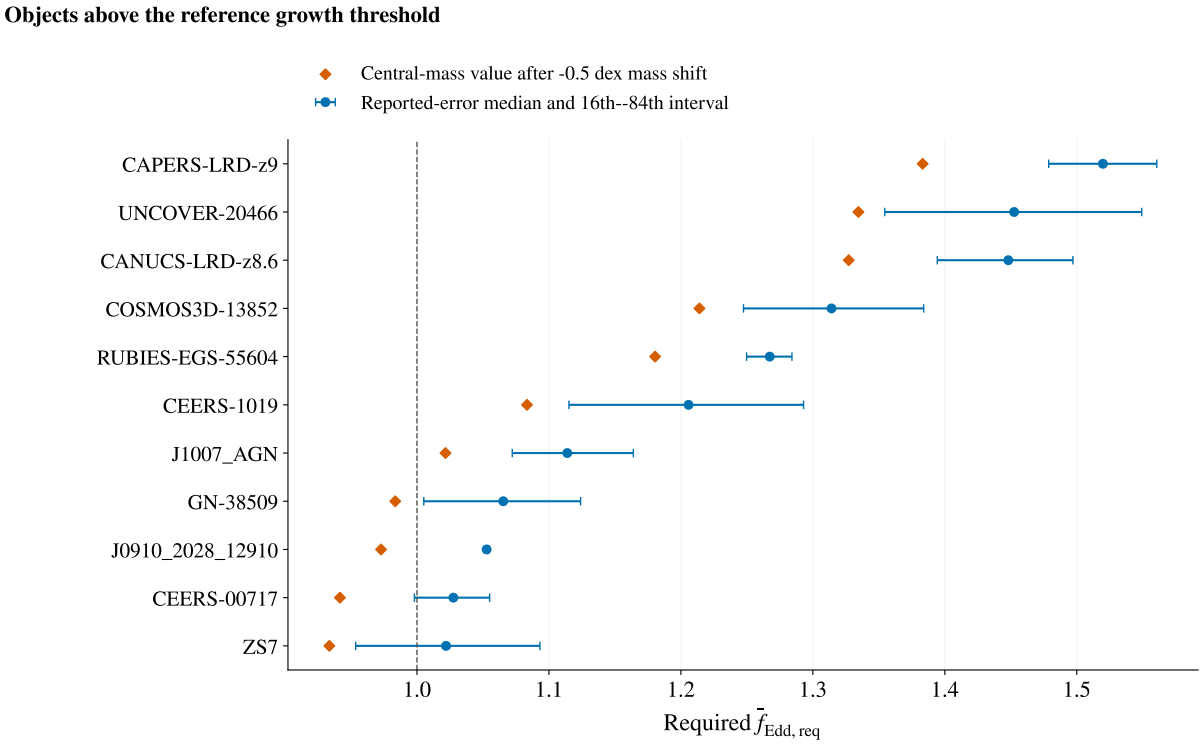


Figure 1: The eleven primary objects with $f_0 > 1$ under the reference model. Circles and bars show medians and 16th–84th percentile intervals of the required lifetime-average Eddington ratio over each object's mass-error draws; diamonds show f_0 after a -0.5 dex mass shift. The vertical line is the reference threshold. The selected objects all have reported errors; these intervals do not include additional estimator systematics.

Downward mass shifts of 0.3 and 0.5 dex reduce the number of primary objects above unity at their central masses from eleven to nine and seven, respectively (Table 3). CAPERS-LRD-z9, CANUCS-LRD-z8.6, UNCOVER-20466, COSMOS3D-13852, and RUBIES-EGS-55604 retain $P \geq 0.95$ at -0.5 dex. Biases specific to each source may exceed this offset (Section 5.1), and the required rates still depend on seed mass, formation time, and radiative efficiency.

The separate calibration test changes the interpretation of J0910_2028_12910: its unshifted probability falls from above 0.999 to 0.745 (Table 4). RUBIES-EGS-55604 retains a probability of 0.979 after both the added scatter and the -0.5 dex shift. These results assume independent Gaussian calibration errors; they do not constrain unquantified biases in the other targets.

Adopting the published Baccus measurements and varying the treatment of unmatched or uncertain identities leaves the threshold counts and five highest-ranked objects unchanged. Alternative measurements for six other objects also leave them below the reference threshold (Appendix A.3).

Table 3: Sensitivity to fixed mass offsets under the reference model. The slash-separated counts are for objects above unity at the central mass, at the 16th percentile, and with conditional $P(\overline{f}_{\rm Edd,req} > 1) \geq 0.95$. The probability-based counts use 211 primary and 226 expanded-sample objects with reported errors; central-mass counts use all 223 and 238 objects.

Mass offset (dex)	Primary	Expanded
−0.5	7 / 5 / 5	10 / 7 / 6
−0.3	9 / 8 / 6	13 / 10 / 7
0	11 / 9 / 8	17 / 13 / 11
+0.3	14 / 14 / 13	21 / 18 / 17
+0.5	18 / 15 / 14	25 / 21 / 18

Table 4: Conditional probabilities of exceeding the reference required average Eddington ratio of 1. The reported-error columns use exact normal tails for these two symmetric-error measurements; the calibration columns add independent 0.5-dex Gaussian scatter. The existing Monte Carlo results and sample membership are unchanged.

	No mass shift		−0.5 dex shift	
Object	Reported	+ calibration	Reported	+ calibration
RUBIES-EGS-55604	> 0.999	0.999	> 0.999	0.979
J0910_2028_12910	> 0.999	0.745	< 0.001	0.367

4.3 Seed mass, growth onset, and radiative efficiency

Increasing the seed mass to $10^3\,M_\odot$ reduces the number of primary objects with required ratios above unity at their central masses from 11 to 5; for $10^5\,M_\odot$ seeds, no objects in either sample exceed 1. Delaying seeding to $z = 20$ raises the primary count to 21. Adopting $\epsilon = 0.05719$, the ideal thin-disc efficiency for a non-spinning black hole ($a = 0$; J. M. Bardeen et al. 1972), leaves no objects in either sample above unity at their central masses (Table 5). Each test changes one reference parameter.

Table 5: Objects requiring $\overline{f}_{\rm Edd,req} > 1$ at their central masses under fixed growth assumptions. Each row changes only the named parameter from the reference ($M_{\rm seed} = 10^2 M_\odot$, $z_{\rm seed} = 30$, $\epsilon = 0.1$, $B_{\rm merge} = 1$). Counts use central mass estimates and provisional object identities.

Scenario	Primary (223)	Expanded (238)
Reference	11	17
$M_{\rm seed} = 10^3 M_\odot$	5	7
$M_{\rm seed} = 10^5 M_\odot$	0	0
$z_{\rm seed} = 20$	21	28
$\epsilon = 0.05719$	0	0

GN-z11 requires an average Eddington ratio of 1.437 under the reference assumptions. Delaying seeding to $z = 20$ raises this to 1.880, above UNCOVER-20466's 1.733, reversing their order. Appendix D shows the dependence on seed mass and starting redshift for both objects.

Figure 2 shows illustrative tracks at four fixed efficiencies. Accretion rates are selected separately in each panel for proximity to the observed masses, so the panels do not isolate the effect of efficiency. Appendix B.5 specifies the display selection.

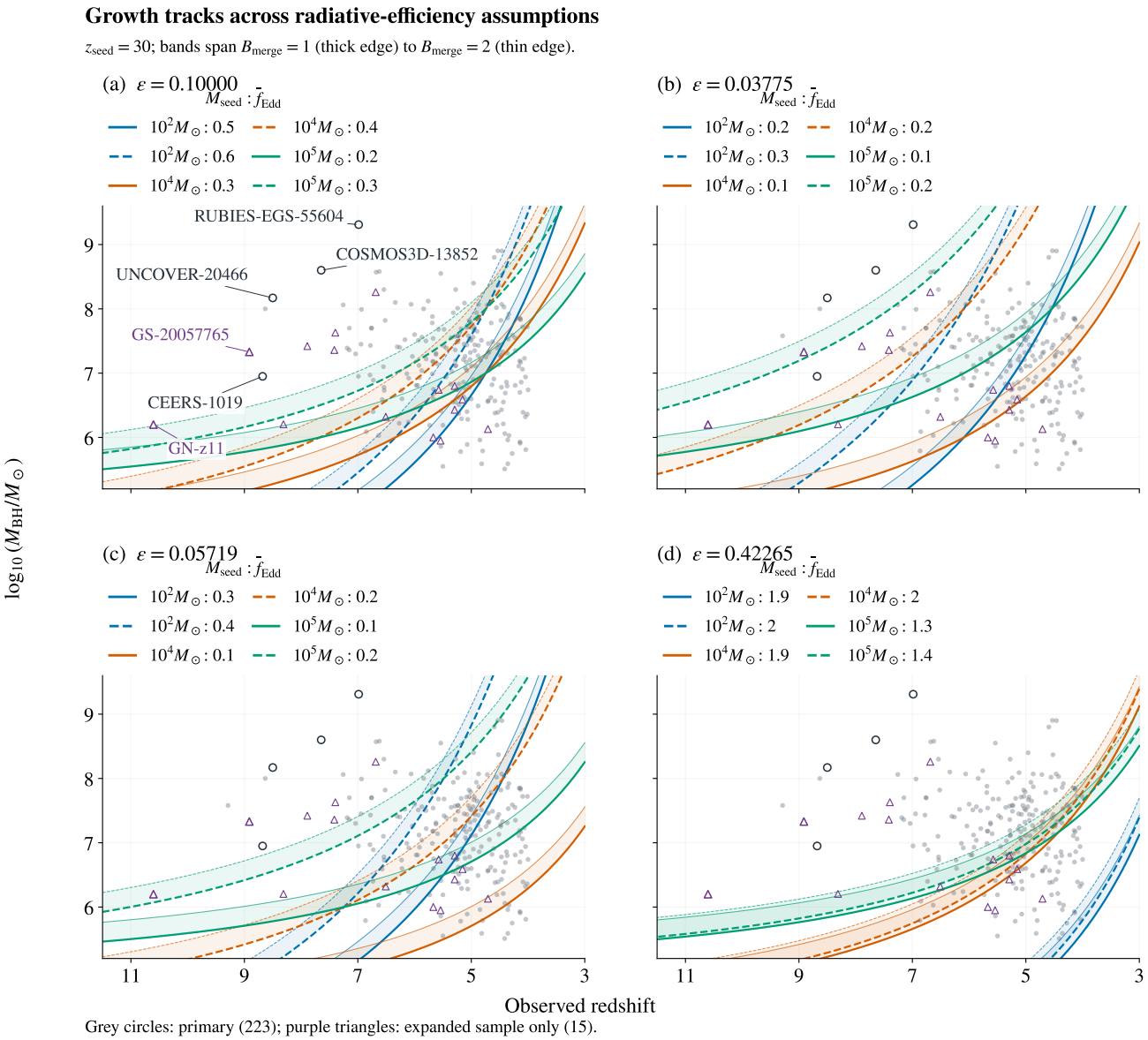


Figure 2: Growth tracks with 223 primary and 15 additional objects. Panels show 4 constant radiative efficiencies at $z_{seed} = 30$; colours distinguish seeds of 10^2, 10^4, and $10^5 M_\odot$, and legends list the average Eddington ratios, selected separately in each panel to lie near the sample. Shaded bands span merger boosts $B_{merge} = 1$–2, with thick and thin edges respectively. Six targets are labelled in the first panel and highlighted in every panel. Objects with unresolved identities or without eligible masses are omitted.

Figure 3 shows the maximum fixed efficiency allowing $\overline{f}_{Edd,req} \leq 1$ for 5 targets as a function of seed mass. The $z_{seed} = 20$ and 30 curves show GN-z11's greater sensitivity to the starting time. These boundaries use central mass estimates and assume no merger boost.

Appendix C gives the fractions of objects whose masses can be reproduced from each seed-mass range under specified accretion rates, spin-dependent efficiencies, and merger contributions.

4.4 External dynamical-mass check: A2744-QSO1

For A2744-QSO1, we compare the virial mass adopted in the catalogue $\log_{10}(M_{BH}/M_\odot) = 7.3 \pm 0.2$ (J. E. Greene et al. 2024) with the independent MOKA3D dynamical mass estimate 7.7 ± 0.3 (I. Juodžbalis et al.

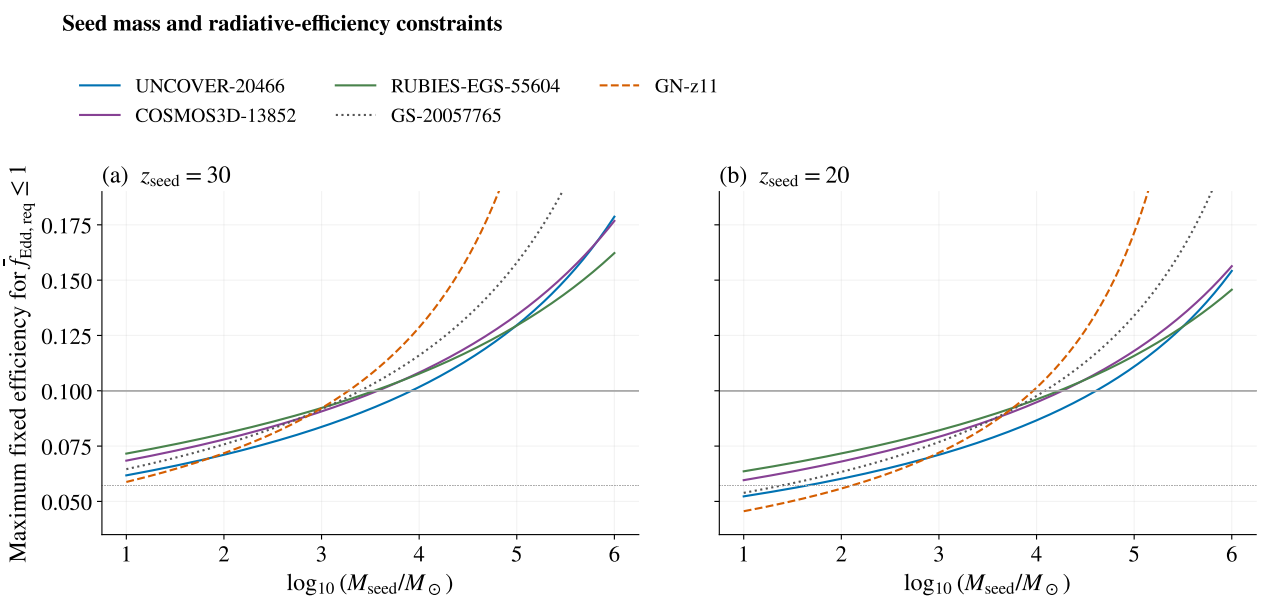


Figure 3: Seed-mass–efficiency boundaries for Eddington-limited average growth at 2 starting redshifts. Each curve uses an adopted central mass and no merger boost. Below a curve the required average is at most unity. Horizontal guides mark $\epsilon = 0.1$ (solid grey) and 0.05719 (dotted grey). GN-z11 (dashed) and GS-20057765 (dotted) belong only to the expanded sample; the other objects are primary. Each point assumes a radiative efficiency that remains constant throughout growth.

2026b), using $z = 7.04$ and the reference model. The dynamical estimate depends on lens reconstruction and inclination. Using symmetric normal approximations to the quoted log-mass errors, the required average Eddington ratio rises from 0.929 to 1.000, with 16th–84th percentile intervals of 0.895–0.964 and 0.947–1.052, respectively. For the dynamical mass, $P(\overline{f}_{\rm Edd,req} > 1) = 0.497$; the unrounded central ratio is just below unity in the reference cosmology (see Appendix B.3 for sensitivity to the cosmic-age approximation). Calibrating the virial sample would require independent measurements for more objects.

4.5 Accretion-onset sensitivity

We compare earlier accretion onset for CAPERS-LRD-z9, UNCOVER-20466, and CANUCS-LRD-z8.6, the three highest-ranked primary objects under the reference growth model, together with GN-z11 as an expanded-sample UV-mass comparison object. We use the growth component of Eq. (1) and Section 2 of P. Dayal (2024), with an optimistic accretion onset at matter–radiation equality ($z_{\rm acc} = 3400$), and separately test delayed onset. The seed mass denotes the mass at the start of accretion. For every onset, including the $z_{\rm acc} = 30$ controls, we use the radiation-inclusive cosmic ages in Eq. 13.

With no mergers and constant radiative efficiency, the minimum onset seed that reaches an adopted mass for $0 \leq \overline{f}_{\rm Edd} \leq f_{\rm cap}$ is

$$\log_{10}\left(\frac{M_{\rm seed,min}}{M_\odot}\right) = \log_{10}\left(\frac{M_{\rm BH}}{M_\odot}\right) - f_{\rm cap}\frac{1-\epsilon}{\epsilon}\frac{t(z_{\rm obs}) - t(z_{\rm acc})}{t_{\rm Edd}\ln 10}. \tag{4}$$

Here $f_{\rm cap}$ is a cap on the lifetime-average luminosity-based Eddington ratio. We evaluate 47 onset redshifts between 20 and 3400, caps of 0.3, 1, and 2, and efficiencies of 0.1 and $1 - \sqrt{8/9} = 0.05719$. Efficiency remains fixed at every cap. Reported asymmetric log-mass errors are propagated with 10,000 equal-side half-normal Monte Carlo draws per object, using stable object-specific streams with seed 20261001.

Redshifts remain fixed; a −0.5-dex mass shift is tested separately on the same draws. The onset grid is recorded in the analysis configuration listed under Data and code availability.

Table 6: Minimum onset seed masses ($M_\odot$) for the four targets at their adopted central masses, $\epsilon = 0.1$, $f_{\rm cap} = 1$, and no mergers. All columns use radiation-inclusive cosmic ages. Values are rounded; Figure 4 shows the reported-error sensitivity.

Object	$z_{\rm acc} = 30$	$z_{\rm acc} = 100$	$z_{\rm acc} = 3400$
CAPERS-LRD-z9	8243	1587	1144
UNCOVER-20466	8527	1642	1184
CANUCS-LRD-z8.6	7334	1412	1018
GN-z11	1924	370	267

At $\epsilon = 0.1$ and $f_{\rm cap} = 1$, equality-onset minimum seeds are approximately 1144, 1184, 1018, and 267 $M_\odot$, respectively (Table 6). Thus an earlier start alone does not allow a 100 $M_\odot$ seed to reach any of these four central masses under these assumptions. Relative to $z_{\rm acc} = 30$, equality onset adds 98.73 Myr and reduces each minimum seed by a factor of 7.20. Moving onset from $z_{\rm acc} = 100$ to 3400 adds only about 16.35 Myr and reduces it by a further factor of 1.39. Lower efficiency or larger allowed average rates reduce the required seeds further. The lower-efficiency control permits 100 $M_\odot$ seeds starting at $z = 30$ to reach all four central masses.

5 Discussion

The sensitivity to radiative efficiency prevents these constraints from uniquely identifying a seed-formation history. A required average below unity permits accretion below the Eddington limit throughout the growth interval; it does not exclude super-Eddington episodes in the actual history.

5.1 Mass-estimator limitations

Some mass-estimator biases may exceed the 0.5 dex reduction tested above (Table 7). X. Lin et al. (2026) discuss possible 1–2 dex virial-mass overestimates from electron scattering. Applying these shifts to COSMOS3D-13852 lowers its required average Eddington ratio to 1.115 and 0.916, respectively, under the reference assumptions. These are sensitivity tests, not measured corrections for this object.

Table 2 gives the reduction in each adopted log mass needed to bring its required average Eddington ratio to 1 under the reference assumptions. For J0910_2028_12910, a reduction of 0.330 dex is sufficient. This is smaller than its separately reported 0.5-dex calibration scale (C. J. Baccus & X. Xu 2026), although its 0.01-dex quoted error gives $P > 0.999$ in the reported-error calculation. Treating the separate calibration scale as independent Gaussian scatter lowers that probability to 0.745 (Table 4). COSMOS3D-13852 requires a reduction of 1.576 dex, within the 1–2 dex range of possible scattering bias. UNCOVER-20466 requires 1.921 dex; its lensing and broad-line assumptions still require independent tests.

Confirming the weak broad Hβ component in CEERS-1019 would strengthen the basis of its virial mass (R. L. Larson et al. 2023), although calibration uncertainty would remain. An independent mass estimate for J0910_2028_12910 could test whether it still requires an average Eddington ratio above 1 under the reference assumptions. Shared calibration errors would limit how strongly agreement between estimators supports the inferred mass.

A mass estimate based on the Balmer lines would test whether GN-z11's growth requirement depends on its UV mass estimator.

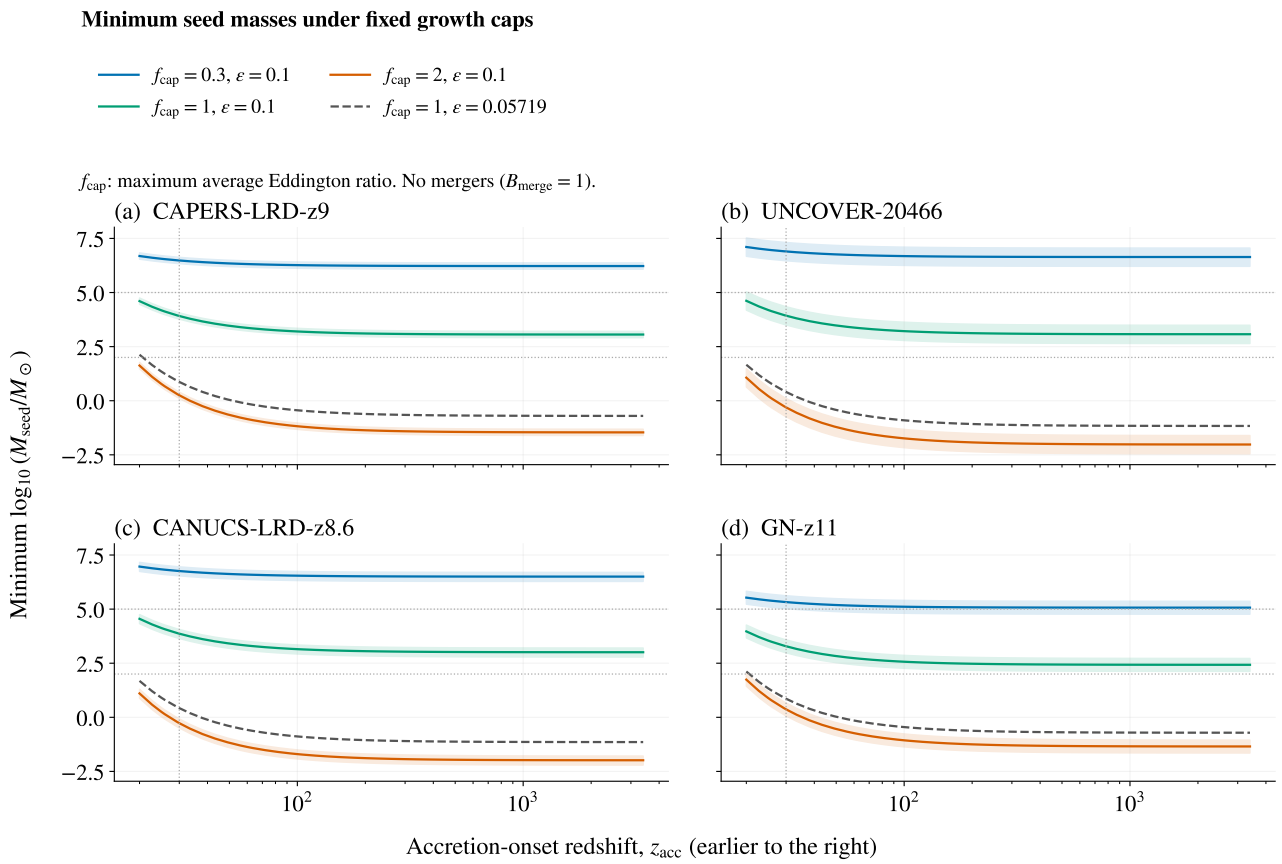


Figure 4: Minimum onset seed mass versus accretion-onset redshift for the four targets. Solid curves vary the average Eddington-ratio cap at fixed $\epsilon = 0.1$; shaded bands propagate the 16th–84th percentiles of reported mass errors. Dashed curves show the lower-efficiency sensitivity. Earlier onset reduces the required mass, with diminishing gains in available time.

5.2 Growth constraints for individual objects

V. Kokorev et al. (2023) compared light and heavy seeds for UNCOVER-20466, while R. Maiolino et al. (2024) discussed heavy seeds and super-Eddington growth from lighter seeds for GN-z11. The reference calculation recovers these growth tensions and quantifies their sensitivity across the catalogue to reported mass errors, systematic mass shifts, and alternative growth assumptions.

UNCOVER-20466 and GS-20057765 require average Eddington ratios above 1 for $10^3 M_\odot$ seeds, but below 1 for $10^5 M_\odot$ seeds, with the other reference assumptions unchanged. At radiative efficiency $\epsilon = 0.05719$, $100\,M_\odot$ seeds formed at $z = 30$ without merger growth require average Eddington ratios of 0.793 and 0.740, respectively. If accretion occurs at the Eddington limit during active phases and ceases between them, these values require activity for about 79% and 74% of the available time. The constraints for GS-20057765 depend on its tentative broad-Hβ component tracing a broad-line region.

Table 7: Mass-estimator limitations for the 11 primary objects requiring $\overline{f}_{\rm Edd,req} > 1$ under the reference assumptions. Caveats summarize the source records cited in Table 2.

Object	Mass-estimator caveat
CAPERS-LRD-z9	Hβ; dense-gas/BLR geometry, correlated dust alternatives; the 0.15 dex error includes calibration coefficients but not the full high-redshift systematic.
CANUCS-LRD-z8.6	Hβ; mass already lens-corrected; alternate lens model and calibration-scatter decomposition remain caveats.
J1007_AGN	Observed total-Hβ luminosity estimator; correlated dust correction raises the mass; separate 0.43 dex calibration scale is not added.
UNCOVER-20466	Hβ; mass depends on lens model and broad-line-region interpretation.
COSMOS3D-13852	Hβ; possible scattering bias; 1–2 dex shifts tested here.
RUBIES-EGS-55604	Hβ; dust-corrected mass; separate 0.5 dex calibration scale.
CEERS-1019	Weak broad Hβ component.
GN-38509	Hα; quoted mass errors include 0.3 dex calibration scatter.
J0910_2028_12910	Hα; 0.01 dex formal error excludes a 0.5 dex calibration scale.
CEERS-00717	Hα; local calibration; no quantitative systematic uncertainty reported.
ZS7	Hβ; quoted 0.4 dex already includes calibration scatter.

5.3 Interpreting primordial-seed growth limits

Equal masses at accretion onset and equal growth parameters give identical growth regardless of seed origin. The $z = 30$ starting time and equality-onset accretion are assumed benchmarks. A primordial black hole (PBH) interpretation requires a seed present before equality; its formation, pre-onset growth, survival, gas supply, abundance, and external observational constraints remain outside this calculation. Section 3 of P. Dayal (2024) additionally models halo assembly, gas, star formation, and feedback for UHZ1 and GHZ9. P. Dayal (2026) compares astrophysical and primordial seeding using models that include gas supply, star formation, feedback, and population predictions. The four-target analysis here isolates the effects of seed mass, accretion onset, and radiative efficiency on the required growth.

5.4 Limitations

The contributing studies have different selection criteria and detection limits, so these sample counts cannot be interpreted as population fractions without completeness corrections. Virial calibration and geometry errors are incompletely reported.

Unresolved object matches limit the accuracy of catalogue counts, but excluding them leaves the threshold results unchanged (Appendix A.3).

Including radiation raises the reference required ratios by 0.14–0.26% without changing the adopted central-mass threshold counts. CEERS-00717's 16th-percentile ratio and the ratio from A2744-QSO1's external dynamical-mass estimate cross unity; both shifts are much smaller than their mass-error intervals (Appendix B.3).

Equation 2 describes growth through an average accretion rate, a fixed merger multiplier, and constant radiative efficiency. Feedback and individual accretion histories remain unspecified. The probabilities hold these assumptions fixed. Averaging over them would require justified parameter distributions and correlations; identifying seed origins would also require formation models and population information.

6 Conclusions

For the adopted masses and growth assumptions, we find:

- For $100\,M_\odot$ seeds formed at $z = 30$, $\epsilon = 0.1$, and no merger contribution, 11 of the 223 objects in the primary sample require an average Eddington ratio above 1. Eight do so in at least 95% of Monte Carlo draws from the adopted mass-error model. Adding the separately reported 0.5-dex calibration scale lowers the probability for J0910_2028_12910 to 0.745. Other probabilities retain the source-reported errors, whose calibration coverage varies.
- At least 95% of draws still require an average Eddington ratio above 1 after reducing the masses by 0.5 dex for five objects: CAPERS-LRD-z9, CANUCS-LRD-z8.6, UNCOVER-20466, COSMOS3D-13852, and RUBIES-EGS-55604. RUBIES-EGS-55604 retains a probability of 0.979 when its separate calibration scale is also included. Unquantified lensing and line-broadening biases remain outside these probabilities.
- Delaying seeding to $z = 20$ makes GN-z11 the highest-ranked object. At constant efficiency $\epsilon = 0.05719$, every central mass in the 238-object expanded sample is reachable from a $100\,M_\odot$ seed at $z = 30$ with an average Eddington ratio below 1.
- For the four selected targets, accretion beginning at matter–radiation equality requires approximately 270–1200 $M_\odot$ seeds at $\epsilon = 0.1$ and an average Eddington ratio capped at 1. At $\epsilon = 0.05719$, $100\,M_\odot$ seeds starting at $z = 30$ reach all four central masses with an average Eddington ratio below 1.
- For A2744-QSO1, replacing the adopted virial mass with the independent dynamical estimate raises the required average Eddington ratio from 0.929 to approximately 1. Its mass uncertainty allows average rates both below and above the Eddington limit.

Data and code availability

Code and data are available at `https://github.com/nniickels/highz-accretion-atlas`. The unchanged two-object calibration calculation is preserved at commit `77908cdaadb5`. That revision predates the expanded catalogue. The expanded catalogue and scientific results correspond to commit `a1bef516db5c`. Catalogue source records are in `data/`, manuscript results in `results/for_manuscript/`, and analysis code/notebooks in `src/` and `scripts/`. The unchanged calibration implementation is `src/internal/calibration_sensitivity.py`; its input and result are `target_robustness.csv` and `calibration_sensitivity.csv` in the manuscript-results tables. Earlier-onset inputs and products are in `data/scenarios/v4_pbh_growth.json` and `results/v4/`. Pinned dependencies and reproduction commands are in `requirements/core.txt` and `docs/guides/reproducibility.md`.

AI assistance. OpenAI models `gpt-5.6-sol`, `gpt-6-astra`, and `gpt-6.1-sol` assisted with scraping published papers for catalogue data and with code review, repository documentation help, and code development struggles through Codex. Appendix A describes the source checks and their coverage. The author is responsible for the analysis, interpretation, and final manuscript.

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Appendix A Catalogue sources, selection, and validation

Figure 5 summarizes the sample accounting; Table 8 lists the contributing source families.

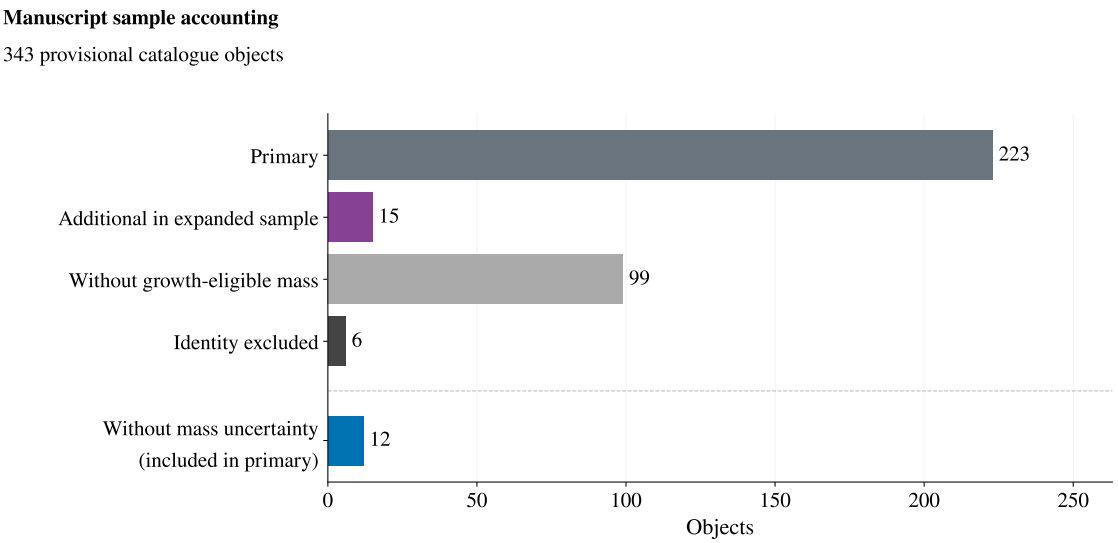


Figure 5: Sample accounting for 343 provisional catalogue objects. The first four bars show mutually exclusive categories: 223 primary objects, 15 additional expanded-sample objects, 99 retained without eligible masses, and 6 identity exclusions. The final bar shows the 12 primary objects without reported mass uncertainties; these are included in the primary count. The expanded sample contains 238 objects. Probability calculations use 211 primary and 226 expanded-sample objects.

Source validation. Independent source records check the central masses and error fields of all 248 numerical mass measurements, including missing errors for 12 measurements. Separate checks cover all 355 central redshifts and 328 available coordinate pairs. The records retain source versions, table locators, and conversion notes. These checks test transcription and unit conversion; they do not validate the mass estimators or resolve ambiguous identities. Other fields, including luminosities, magnifications, and evidence classifications, have partial independent coverage. The fixtures and verification commands are in `data/validation/`.

Released tables. In `data/processed/v3/`, `v3_accreting_measurements.csv` contains the measurement records, and `v3_accreting_objects.csv` contains one adopted record per provisional object. Join them using `measurement_id` and `physical_object_id`; `host_system_id` links systems assigned to the same host. The measurement table records the adopted choice in `preferred_measurement_flag` and its reason. Standardized black-hole masses and their errors are in $\log_{10}(M_{\rm BH}/M_\odot)$ and dex; redshift is dimensionless and coordinates are in degrees. Empty numeric cells denote missing values, not zeros. Limits and conditional masses have separate flags. The file `results/for_manuscript/tables/publication_object_selection.csv` provides `publication_primary_flag` and `publication_exploratory_flag`; the latter selects the expanded sample, including the primary objects. Source versions and locators remain in the measurement table and `data/source_provenance_registry.csv`.

Auxiliary coordinates are attributed in the registry to THRILS (T. A. Hutchison et al. 2025), UHZ1 spectroscopy (A. D. Goulding et al. 2023), and the versioned JADES DR3 GOODS-S prism catalogue.

Table 8: Source-family inventory. Counts are catalogue measurements / growth-eligible measurements / primary-sample objects (M/E/P). The columns sum to 355/248/223; E includes alternatives, whereas P counts each selected object once. "Balmer" denotes single-epoch virial masses. "Context" denotes mass estimates retained for discussion but excluded from growth calculations because of their method or interpretation. "Upper limit" denotes an upper bound rather than a measured central mass; these bounds are also excluded from growth calculations. EELG: extreme emission-line galaxy; SED: spectral energy distribution; BLR: broad-line region; LRD: little red dot; WFSS: wide-field slitless spectroscopy.

Source	M/E/P	Mass	Selection channel; principal caveat
Baccus C. J. Baccus & X. Xu (2026)	49/49/48	Balmer	Frozen preprint masses; calibration
Chavez Ortiz Ó. A. Chávez Ortiz et al. (2026)	1/0/0	Context	Model-dependent mass
Chisholm J. Chisholm et al. (2024)	1/0/0	—	No canonical mass
Davis / THRILS K. Davis et al. (2026)	6/6/6	Balmer	Virial calibration
Fei / GLIMPSE Q. Fei et al. (2026)	10/10/8	Balmer	Lensing corrections
Greene / UNCOVER J. E. Greene et al. (2024)	9/9/9	Balmer	Lens model; BLR interpretation
Harikane Y. Harikane et al. (2023)	10/10/5	Balmer	Local virial calibration
Juodzbalis / JADES I. Juodžbalis et al. (2026a)	23/23/17	Balmer	Extinction; calibration
Killi M. Killi et al. (2024)	1/1/1	Balmer	Linear mass-error conversion
Kocevski / LRDs D. D. Kocevski et al. (2025)	6/6/6	Balmer	Dust correction; calibration
Larson / CEERS-1019 R. L. Larson et al. (2023)	1/1/1	Balmer	Broad-component significance
Lin / ASPIRE X. Lin et al. (2024)	16/16/16	Balmer	Virial calibration
Lin / COSMOS-3D X. Lin et al. (2026)	13/13/13	Balmer	Possible scattering bias
Lyu / SMILES J. Lyu et al. (2024)	19/0/0	—	SED-dependent AGN identification
Maiolino / GN-z11 R. Maiolino et al. (2024)	1/1/0	UV	UV estimator outside primary
Mascia S. Mascia et al. (2026)	8/0/0	—	No canonical mass
Matthee / EIGER-FRESCO J. Matthee et al. (2024)	20/20/19	Balmer	Dust; virial calibration
Mazzolari G. Mazzolari et al. (2025)	25/0/0	—	Star-formation overlap
Leung / MEOW G. C. K. Leung et al. (2026)	12/0/0	—	SED-dependent AGN fraction
Naidu / MoM-BH*-1 R. P. Naidu et al. (2026)	1/0/0	Context	Scattering; stellar alternative
Napolitano / GHZ9 L. Napolitano et al. (2025a)	1/0/0	Context	Mass assumes Eddington rate
Napolitano / GHZ4,7 L. Napolitano et al. (2025b)	2/0/0	—	Tentative diagnostics
Ren / ALPINE-CRISTAL W. Ren et al. (2025)	7/7/1	Balmer	Conditional BLR interpretations

Table 8 continued

Source	M/E/P	Mass	Selection channel; principal caveat
Scholtz / JADES J. Scholtz et al. (2025)	21/0/0	—	Tentative classifications
Skyfire / CEERS D. D. Kocevski et al. (2026)	22/22/22	Balmer	Virial calibration
Tang M. Tang et al. (2025)	3/0/0	—	Ionizing-source ambiguity
Taylor / CEERS-RUBIES A. J. Taylor et al. (2025)	37/37/35	Balmer	Slit loss; dust; calibration
Treiber / UNCOVER H. Treiber et al. (2025)	2/0/0	—	Stellar alternatives
Ubler / ZS7 H. Übler et al. (2024)	1/1/1	Balmer	Error includes calibration scatter
Bogdan / Zou / UHZ1 Á. Bogdán et al. (2024); F. Zou et al. (2026)	2/0/0	Context	Later detection disputed
Zhang / LRDs Z. Zhang et al. (2026)	5/0/0	Upper limit	Mass upper limits
Zhuang / NEXUS M.-Y. Zhuang et al. (2026)	15/12/12	Balmer	Twelve masses lack errors
Taylor / CAPERS Taylor et al. (2025)	1/1/1	Balmer	prism resolution; gas enshrouded lrd; dust alternatives correlated; high redshift virial calibration; coordinate frame unreported
Tripodi / CANUCS Tripodi et al. (2025)	1/1/1	Balmer	high redshift virial calibration; source corrected lensing; mass dust treatment unspecified; morishita lens model differs
Schindler / J1007 Schindler et al. (2025)	1/1/1	Balmer	high redshift virial calibration; dust alternative correlated; total hbeta luminosity recipe; redshift error text table discrepancy
Takechi / Y1 Takechi et al. (2026)	1/1/0	Balmer	weak broad hbeta; outflow ambiguity; mass interval confidence unspecified; calibration scatter inclusion unspecified; host centroid not nucleus; source corrected lensing
Fudamoto / SAPPHIRES Fudamoto et al. (2025)	1/0/0	—	no published mbh; weak hbeta peak; foreground neighbour identity unresolved; compactness contaminated; no broad narrow decomposition

A.1 Expanded-sample membership

Seven masses are explicitly flagged as conditional on a broad-line-region interpretation: 6 candidates with intermediate-width Hα emission and possible outflow contamination (W. Ren et al. 2025), and RUBIES-EGS-49140 (A. J. Taylor et al. 2025), for which an alternative non-AGN interpretation is recorded. These 7 objects enter the expanded sample only.

Four further JADES objects—GS-20057765, GS-20030333, GS-164055, and GN-4685—enter the expanded sample only because their individual broad-Hβ detections are tentative, although broad emission is supported by the stacked spectrum (I. Juodžbalis et al. 2026a).

GLIMPSE-38548 and GLIMPSE-7404 also enter the expanded sample only; the catalogue classifies them as tentative broad-Hα AGN (Q. Fei et al. 2026).

A.2 Adopted measurements and uncertain object matches

The catalogue fields `evidence_status` and `evidence_status_basis` record the evidence classification and its source-based rationale. For three pairs of entries, it remains unclear whether both refer to the same object: Baccus GDS_1210_9515 and JADES GS-8083; GS-10013704 and Scholtz 99671; and Scholtz 16745 and 208643. All six entries are excluded from the analysis. For CEERS-2782, we adopt the higher-signal-to-noise RUBIES-EGS-50052 spectrum preferred by the source study; for NX10835, we adopt the newly available NEXUS mass estimate.

A.3 Sensitivity to measurement choice and sample membership

In the expanded sample, 17 objects exceed unity at their central masses, 13 exceed unity at the 16th percentile, and eleven have $P \geq 0.95$; 19 exceed unity at the 84th percentile. These statistics use 226 objects with reported errors.

The baseline catalogue retains the Baccus & Xu measurements from `arXiv:2512.03281v1` to preserve the frozen catalogue's membership and adopted measurements. The bibliography cites the published article; the measurement records identify the preprint version supplying the baseline values. Replacing those values is tested separately below. A full update would also require reviewing the published sample's membership and classifications. We matched 44 catalogue objects to the published sample using identifiers and sky positions. For this sensitivity test, we replaced their preprint values with the redshifts, black-hole masses, and mass uncertainties reported in Table 1 of C. J. Baccus & X. Xu (2026). The threshold counts and the five highest-ranked objects remain unchanged in both samples (Table 9). Of five unmatched records, one is excluded for uncertain identity; retaining or omitting the other four leaves these results unchanged. Omitting them leaves 219 primary and 234 expanded-sample objects.

Table 9: Effect of replacing the adopted Baccus preprint values with published measurements, with unresolved identities excluded throughout. Counts denote required ratios above unity at the central masses, 16th percentiles above unity, and conditional $P \geq 0.95$, respectively. The adopted catalogue values define the baseline.

Sample	Measurement treatment	Objects	Counts
Primary	Adopted catalogue values	223	11/9/8
Primary	Published; retain unmatched	223	11/9/8
Primary	Published; omit unmatched	219	11/9/8
Expanded	Adopted catalogue values	238	17/13/11
Expanded	Published; retain unmatched	238	17/13/11
Expanded	Published; omit unmatched	234	17/13/11

Excluding the three uncertain-identity records with eligible masses reduces the primary sample from 226 to 223 objects and the expanded sample from 241 to 238 objects, while preserving threshold counts and top-five rankings in every scenario in Table 5, including the reference probability counts. The median required lifetime-average Eddington ratio across primary objects, evaluated using their central masses, changes from 0.57420 to 0.57425; the objects above the reference threshold remain the same.

Seven alternative-measurement comparisons across 6 objects (2 for CEERS-2782) span mass shifts of -0.600 to $+0.949$ dex and shifts in $\overline{f}_{\mathrm{Edd,req}}$ of -0.083 to $+0.121$ (Fig. 6). Both required Eddington ratios remain below unity in every pair. Testing the objects above the threshold requires additional independent mass estimates.

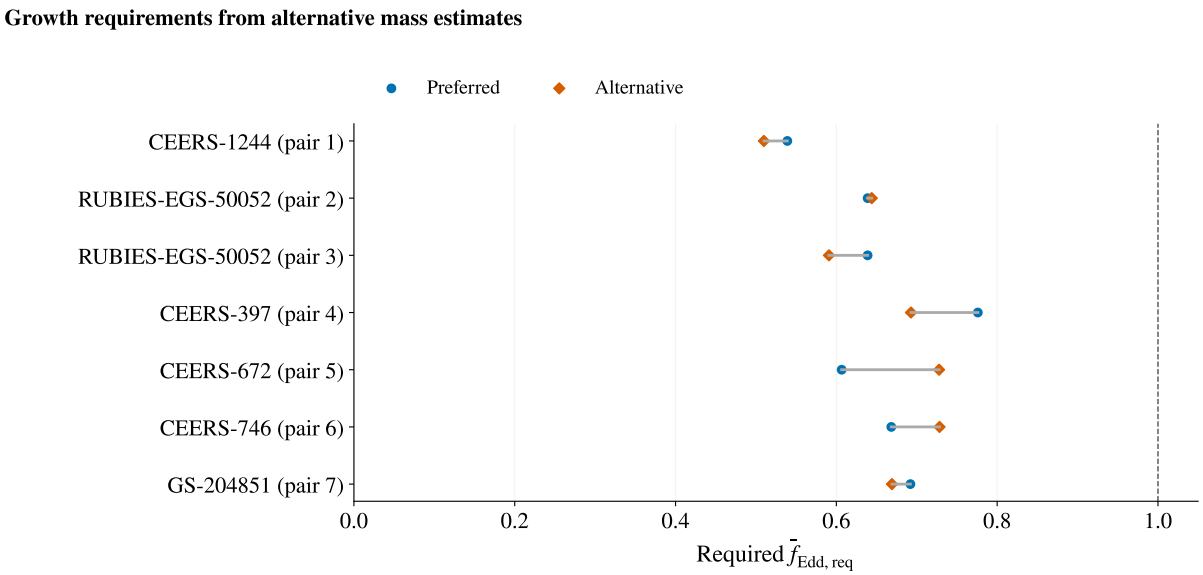


Figure 6: Preferred (circles) and alternative (diamonds) required ratios at the central masses for 7 measurement pairs across 6 primary objects. Pair numbers follow the released alternative-measurement table, which supplies stable measurement IDs; RUBIES-EGS-50052 (CEERS-2782) has 2 alternatives. All pairs remain below the reference threshold.

Appendix B Growth-model derivations and implementation

B.1 Accretion and exponential growth

For electron scattering in ionized hydrogen, the Eddington luminosity, Eddington ratio, and Eddington timescale are

$$L_{\mathrm{Edd}} = \frac{4\pi G M_{\mathrm{BH}} m_p c}{\sigma_T}, \qquad f_{\mathrm{Edd}} = \frac{L_{\mathrm{bol}}}{L_{\mathrm{Edd}}}, \qquad t_{\mathrm{Edd}} = \frac{M_{\mathrm{BH}} c^2}{L_{\mathrm{Edd}}} = \frac{c\sigma_T}{4\pi G m_p} \simeq 0.45 \text{ Gyr}. \tag{5}$$

Here G, c, m_p, and σ_T denote the gravitational constant, speed of light, proton mass, and Thomson cross section. With radiative efficiency $0 < \epsilon < 1$ and rest-mass inflow rate $\dot{M}_{\mathrm{in}}$,

$$L_{\mathrm{bol}} = \epsilon \dot{M}_{\mathrm{in}} c^2, \qquad \dot{M}_{\mathrm{BH}} = (1-\epsilon)\dot{M}_{\mathrm{in}} = \frac{1-\epsilon}{\epsilon}\frac{f_{\mathrm{Edd}}}{t_{\mathrm{Edd}}} M_{\mathrm{BH}}. \tag{6}$$

The inflow rate is defined at the radiating flow; earlier mass loss is omitted. Dividing Eq. 6 by M_{BH} and integrating at fixed ϵ gives

$$\ln\left(\frac{M_{\mathrm{BH}}}{B_{\mathrm{merge}} M_{\mathrm{seed}}}\right) = \frac{1-\epsilon}{\epsilon}\frac{1}{t_{\mathrm{Edd}}}\int_{t(z_{\mathrm{seed}})}^{t(z_{\mathrm{obs}})} f_{\mathrm{Edd}}(t)\, dt = \frac{1-\epsilon}{\epsilon}\frac{\overline{f}_{\mathrm{Edd}}\,\Delta t}{t_{\mathrm{Edd}}}. \tag{7}$$

Exponentiating gives Eq. 2; solving for the average rate, with a floor of zero, gives Eq. 3. For a constant positive f_{Edd}, the e-folding time is

$$t_{\mathrm{efold}} = \frac{\epsilon}{1-\epsilon}\frac{t_{\mathrm{Edd}}}{f_{\mathrm{Edd}}}. \tag{8}$$

It is 0.05 Gyr at $\epsilon = 0.1$ and $f_{\mathrm{Edd}} = 1$.

B.2 Merger multiplier and inverse seed mass

The implementation evaluates Eq. 2 in log mass:

$$\log_{10}\left(\frac{M_{\rm BH}}{M_\odot}\right) = \log_{10}\left(\frac{M_{\rm seed}}{M_\odot}\right) + \log_{10} B_{\rm merge} + \frac{1-\epsilon}{\epsilon}\frac{\overline{f}_{\rm Edd}\,\Delta t}{t_{\rm Edd}\ln 10}. \tag{9}$$

We evaluate $B_{\rm merge} = 1$ (no boost) and 2 (a factor-of-two mass boost). At fixed accretion history,

$$\frac{M_{\rm BH}(B_{\rm merge}=2)}{M_{\rm BH}(B_{\rm merge}=1)} = 2, \qquad \Delta \log_{10} M_{\rm BH} = \log_{10} 2 \simeq 0.3010. \tag{10}$$

Solving the same relation for seed mass gives

$$\log_{10}\left(\frac{M_{\rm seed,req}}{M_\odot}\right) = \log_{10}\left(\frac{M_{\rm BH}}{M_\odot}\right) - \log_{10} B_{\rm merge} - \frac{1-\epsilon}{\epsilon}\frac{\overline{f}_{\rm Edd}\,\Delta t}{t_{\rm Edd}\ln 10}. \tag{11}$$

Here $M_{\rm seed,req}$ is the seed mass required for the specified average Eddington ratio, radiative efficiency, growth interval, and merger multiplier.

B.3 Cosmic time

Cosmic ages use the flat matter-plus-Lambda relation

$$t(z) = \frac{2}{3H_0\sqrt{\Omega_\Lambda}}\,\mathrm{asinh}\left[\sqrt{\frac{\Omega_\Lambda}{\Omega_m}}(1+z)^{-3/2}\right], \tag{12}$$

with $H_0 = 67.3\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_m = 0.315$, and $\Omega_\Lambda = 0.685$. The Hubble constant is converted to inverse time; ages and growth intervals are in Gyr. Radiation is neglected.

For the radiation check, we calculate cosmic ages numerically as

$$t(z) = \frac{1}{H_0}\int_0^{1/(1+z)} \frac{x\,dx}{\sqrt{\Omega_r + \Omega_m x + \Omega_\Lambda x^4}}, \tag{13}$$

where x is the scale factor. We retain the same H_0 and Ω_m, adopt $\Omega_r = \Omega_m/(1+3400) \simeq 9.262\times10^{-5}$, and set $\Omega_\Lambda = 1-\Omega_m-\Omega_r$ to preserve flatness. Including radiation raises the reference required ratios by 0.14–0.26%, preserving the central-mass counts of eleven primary and 17 expanded-sample objects above unity. Two boundary cases change: CEERS-00717's sampled 16th percentile rises from 0.99805 to 1.00003, and A2744-QSO1's dynamical-mass requirement rises from 0.9996 to 1.0016. Both shifts are much smaller than the uncertainty intervals from the mass-error draws.

B.4 Two-state duty cycle

For a two-state history with the same efficiency in both states,

$$\overline{f}_{\rm Edd} = D f_{\rm burst} + (1-D) f_{\rm quiet}, \qquad D_{\rm req} = \frac{\overline{f}_{\rm Edd,req} - f_{\rm quiet}}{f_{\rm burst} - f_{\rm quiet}}. \tag{14}$$

Here D is the fraction of time spent active, with $f_{\rm burst} > f_{\rm quiet} \geq 0$. The expression applies when $\overline{f}_{\rm Edd,req} \geq f_{\rm quiet}$. Values $D_{\rm req} > 1$ identify histories that cannot reach the observed mass at the assumed rates.

B.5 Mass uncertainties and growth-track plotting

Sampling uses random seed 20260808 and a stable random stream for each object identifier. With 10,000 draws, the Monte Carlo standard error of a probability is at most 0.005, or about 0.0022 at a probability of 0.95; this reflects the finite number of draws. The 12 NEXUS objects without reported mass errors are excluded from probability calculations and uncertainty rankings. Table 2 distinguishes separate calibration scales from scatter already included in the quoted errors.

The mass-scale tests add the same fixed offset to every original mass draw. The final published Killi mass $8^{+0.5}_{-0.4} \times 10^8 M_\odot$ is converted by transforming both linear bounds, giving $\log_{10}(M_{\rm BH}/M_\odot) = 8.90309^{+0.02633}_{-0.02228}$.

For Figure 2, we test average Eddington ratios from 0.1 to 2.0 in steps of 0.1. At each seed mass and fixed radiative efficiency, we plot the 2 rates with the most primary objects within 0.5 dex of either merger prediction ($B_{\rm merge} = 1$ or 2), requiring at least 5 objects. Ties favour smaller median absolute residuals, then smaller rates. Most selected objects lie at $z \simeq 4$–6. This selection is for display; conclusions use the required-rate ranking. Reference rankings and seed-mass–redshift maps use $\epsilon = 0.1$.

Appendix C Compatibility across seed masses and accretion rates

C.1 Efficiency prescription

The maps use the ideal Kerr thin-disc radiative efficiency at the innermost stable circular orbit (ISCO; J. M. Bardeen et al. 1972):

$$\epsilon_a = 1 - \sqrt{1 - \frac{2}{3 r_{\rm ISCO}}}, \tag{15}$$

$$r_{\rm ISCO} = 3 + Z_2 - {\rm sgn}(a)\sqrt{(3 - Z_1)(3 + Z_1 + 2Z_2)}, \tag{16}$$

$$Z_1 = 1 + (1 - a^2)^{1/3}\left[(1 + a)^{1/3} + (1 - a)^{1/3}\right], \qquad Z_2 = \sqrt{3a^2 + Z_1^2}. \tag{17}$$

Here $a = cJ/(GM_{\rm BH}^2)$ is the dimensionless spin, J is the signed angular momentum relative to the disc, and $r_{\rm ISCO}$ is measured in units of $GM_{\rm BH}/c^2$. Neglecting photon capture and stress at the ISCO gives $\epsilon_a = 0.03775$, 0.05719, and 0.42265 for $a = -1, 0, +1$.

For supercritical inflow, we adopt a logarithmic luminosity relation motivated by advection and wind models (J. Poutanen et al. 2007), with coefficient 1 and no radial mass loss:

$$f_{\rm Edd} = 1 + \ln \dot{m}, \qquad \dot{m} = \frac{\dot{M}_{\rm in}}{L_{\rm Edd}/(\epsilon_a c^2)}, \qquad \epsilon_{\rm eff} = \begin{cases} \epsilon_a, & f_{\rm Edd} \le 1, \\ \epsilon_a f_{\rm Edd} \exp(1 - f_{\rm Edd}), & f_{\rm Edd} > 1. \end{cases} \tag{18}$$

The logarithmic relation applies only for $f_{\rm Edd} > 1$. Each cell in Figure 7 holds spin, Eddington ratio, and effective radiative efficiency constant, so $f_{\rm Edd} = \overline{f}_{\rm Edd}$; relativistic slim-disc dynamics are omitted. If efficiency varies, growth depends on $\int[(1 - \epsilon(t))/\epsilon(t)] f_{\rm Edd}(t)\, dt$ and requires both histories. A primordial interpretation requires a separate seed-formation model.

C.2 Seed-mass ranges and compatible fractions

We solve Eq. 11 for each object's required seed mass and check whether it lies within each of four intervals, including the endpoints: 10–100 $M_\odot$ (light), 10^3–10^4 $M_\odot$ (intermediate), 10^4–10^6 $M_\odot$ (heavy), and 10^2–10^6 $M_\odot$ (broad). The grid uses $z_{\rm seed} = 30$, spins $a = -1, 0, +1$, merger multipliers $B_{\rm merge} = 1, 2$, and average Eddington ratios 0.3, 1, and 2, with the efficiency prescription in Section C.1.

The intervals overlap and carry no probability weights; the broad interval covers several possible seed-formation channels. A required seed below an interval's lower bound is incompatible: even its smallest seed would grow beyond the observed mass under those assumptions. We report the percentage of objects whose required seeds fall within each interval, using central mass estimates without propagating their errors. These percentages describe the literature sample. Population fractions would require the contributing studies' selection and detection probabilities.

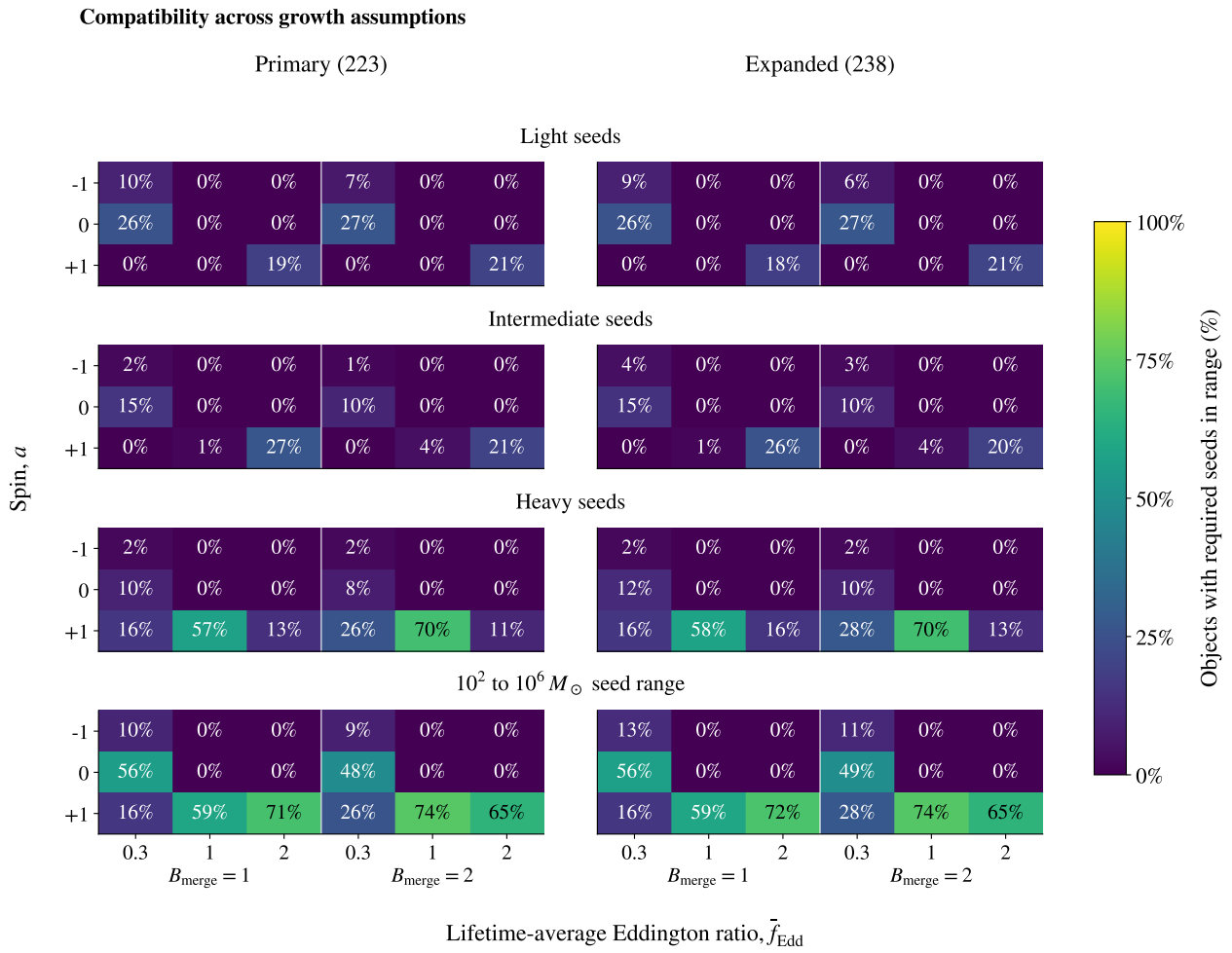


Figure 7: Constant-state compatibility fractions in the primary sample (left) and the expanded sample (right). The expanded sample includes all primary objects. Each column specifies merger multiplier B_{merge} and fixed luminosity ratio f_{Edd}; rows specify ideal spin-dependent efficiency, with the supercritical prescription in Section C.1. Seed intervals overlap and have no assigned probability weights.

The two samples show similar patterns (Fig. 7). For heavy seeds at $a = +1$ and $f_{\mathrm{Edd}} = 1$, changing B_{merge} from 1 to 2 raises compatibility from 57 to 70 per cent in the primary sample and from 58 to 70 per cent in the expanded sample. Higher accretion rates can reduce compatibility when even the smallest seed grows beyond the observed mass.

Appendix D Seed mass and starting redshift

Figure 8 extends the two starting redshifts in Figure 3 to $11 \leq z_{\rm seed} \leq 30$ for UNCOVER-20466 and GN-z11. Later growth onset increases the required average Eddington ratio; larger seeds reduce it. Maps for all objects are available in the repository under `results/v3/parameter_maps/seedredshift_mass_maps/`.

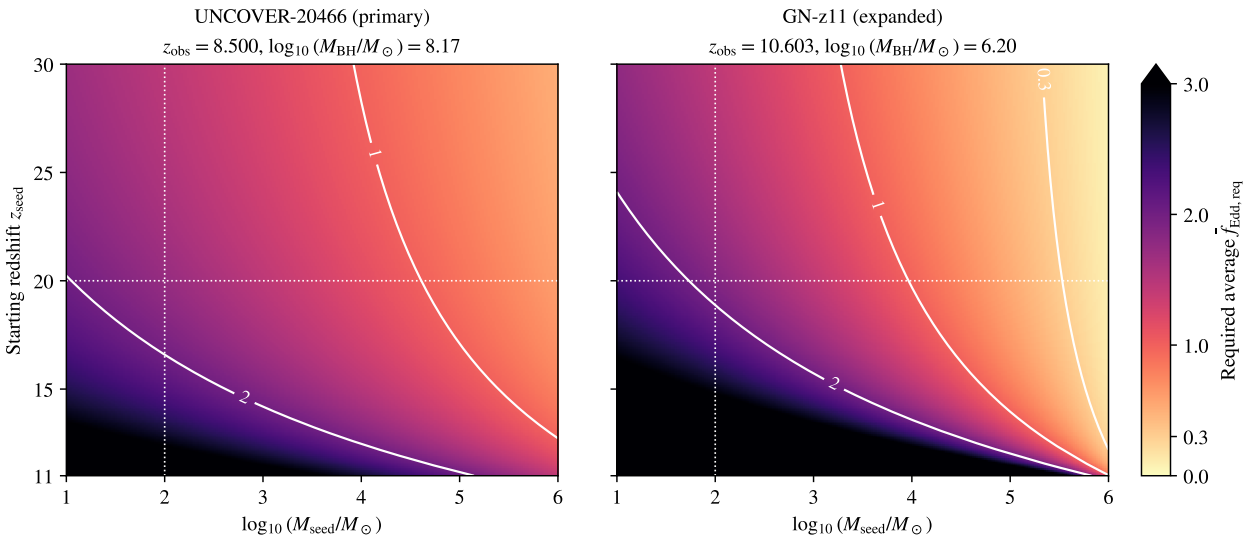


Figure 8: Required average Eddington ratio for UNCOVER-20466 (primary sample) and GN-z11 (expanded sample), using their adopted central masses and redshifts shown above each panel. Both panels use $\epsilon = 0.1$, no merger boost, and the same axes and colour scale. White contours mark required ratios of 0.3, 1, and 2; dotted guides mark a $100\,M_\odot$ seed and $z_{\rm seed} = 20$. Colours saturate above a required ratio of 3. Mass uncertainties are not propagated in these maps.